

Foundations of Deep Learning

Lecture 00

INTRODUCTION

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Section 1

PREREQUISITES & CLASS ORGANIZATION

Schedule

Weekly schedule

- ▶ Wednesday, 12.15-14.00 Spiegelgasse 5, Seminarraum 05.002
- ▶ Thursday, 16.15-18.00 Spiegelgasse 5, Seminarraum 05.001

Teaching assistants

- ▶ Enea Monzio Compagnoni
- ▶ Jim Zhao

Continuous assessment

Note the following split:

- ▶ 15% continuous assessment (short exercises and Q&As given in class)
- ▶ 25% project (writeup and presentation)
- ▶ 25% mid-term exam (end of October during exercise session)
- ▶ 35% final exam

Exercise session

Before the exercise session: You should try to solve the exercises on your own before the exercise session (exercise sheets will be posted weekly on ADAM).

During the exercise session: The responsible TA will explain how to solve the exercises.

Note about the exam: The written exams will be based on similar pen-and-paper exercises.

Project

Projects have to be completed within a group of 2-3 students.
Two types of projects you can pursue are:

1. A novel and sound solution to a problem discussed in the class
2. Thorough theoretical analysis of an existing deep learning approach discussed in the class

The detailed course project guideline can be found in `project_guideline.pdf` in ADAM. Students cannot use their own published work as the course project.

Date to hand in: February 5, 2026.

Class organization

Questions Ask on ADAM (anonymously if you wish).

Material Slides and lecture notes available on ADAM.

The lecture will rely both on slides and hand-written derivations. All the information is in the lecture notes, except for Lectures 0, 1 and 6 (slides only).

The class has a **theoretical flavor**, and we will discuss many proofs/mathematical derivations. The goal of the class is to give you some insights into the inner workings of deep learning.

Prerequisites

1. Machine Learning (classification, regression, kernels, etc)
2. Linear algebra
3. Calculus, Basic concepts in topology
4. Probability theory (random variable, expectation, density, etc)
5. Coding in python (non-mandatory programming exercises but recommended)

Read the lecture notes to make sure you understanding these basic concepts!

Video recordings also available, see the link on Github

Notation

- ▶ Bold small letter $\mathbf{x}$ is a column vector
- ▶ Bold capital letter $\mathbf{A}$ is a matrix
- ▶ $\mathbf{x}^\top, \mathbf{A}^\top$: transpose of a vector and matrix
- ▶ $\mathbb{R}$: reals
- ▶ $a := b$: a is defined by b
- ▶ $\frac{\partial f}{\partial x}$: partial derivative of a function f with respect to x
- ▶ $\frac{df}{dx}$: total derivative of a function f with respect to x
- ▶ ∇ : gradient

Example: Gradient of a Vector-Valued Function

Consider the function $f(\mathbf{x}) = \mathbf{A}\mathbf{x} \in \mathbb{R}^p$, where $\mathbf{x} \in \mathbb{R}^d$, $\mathbf{A} \in \mathbb{R}^{p \times d}$. Our goal is to compute the gradient $\frac{df}{d\mathbf{x}}$.

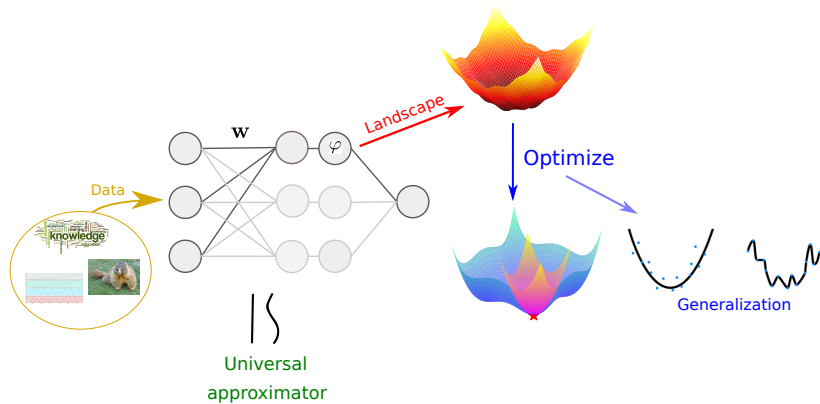
First, note that the dimension of the gradient $\frac{df}{d\mathbf{x}}$ is $\mathbb{R}^{p \times d}$. Let's compute the partial derivative of f w.r.t. a single x_j . We have

$$f_i(\mathbf{x}) = \sum_{j=1}^d A_{ij}x_j \implies \frac{\partial f_i}{\partial x_j} = A_{ij}.$$

Collecting all the partial derivatives in the Jacobian, we obtain the following expression for the gradient:

$$\frac{df}{d\mathbf{x}} = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_p}{\partial x_1} & \cdots & \frac{\partial f_p}{\partial x_d} \end{pmatrix} = \begin{pmatrix} A_{11} & \cdots & A_{1d} \\ \vdots & \ddots & \vdots \\ A_{p1} & \cdots & A_{pd} \end{pmatrix} = \mathbf{A} \in \mathbb{R}^{p \times d}.$$

Syllabus



Syllabus

1. Linear networks, non-linear networks
2. Approximation Theory
3. Complexity Theory
4. Optimization (e.g. stochastic gradient descent)
5. Architecture
6. Optimization Landscape of Neural Networks
7. Neural Tangent Kernel
8. Regularization
9. Generalization bounds
10. Adversarial examples
11. Double Descent